

Blockchain Impacts on Global Supply Chain Operational and Managerial Business Value Processes

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Abstract—The purpose of this article is to investigate the impact of blockchain on global supply chain operational and managerial processes. This article reviews and classifies the existing empirical evidence of blockchain applications in the global supply chain. We utilize the Mooney et al. framework to organize this evidence into operational and managerial business value processes and the effects—automational, informational, and transformational. Our findings indicate that blockchain is not following the Mooney et al. stage model proposed in 1996 since the empirical evidence is evenly spread between the three effect categories. In addition, we have identified possible reasons why blockchain is not following the Mooney et al. stage model. Our study concludes with nine propositions as a guide on how to facilitate blockchain deployment and future research. Limitations of this article concerning blockchain are the lack of consistency and understanding of standards and the lack of information on blockchain implementation, which is referred to as the “grey space.” Blockchain today is where radio-frequency identification (RFID) was in 2002 and the bar code was in 1980. Therefore, future research work could be expanded to investigate case studies of blockchain implementations as they become available across the global supply chain. For practitioners, the empirical evidence presented in this article can help identify applications and implementation steps for building the business case where blockchain can have a greater impact on the organization. This article fills a void in the literature by providing practitioners and academics with a better understanding of blockchain and its applications for implementation.

Key words: Blockchain, global supply chain management, performance measurement

1. INTRODUCTION

THE term “blockchain” was first introduced by Satoshi Nakamoto in 2008 in a study on peer-to-peer (P2P) electronic cash systems [Nakamoto, 2008]. Since then, blockchain has received increased attention from practitioners and academics as an emerging technology [Saber et al., 2019a] and is likely to transform and advance the circular economy [Kouhizadeh et al., 2020]. When blockchain was first introduced, it was

difficult to define, and it was easier to describe blockchain in terms of what it is not, rather than what it could do. It was simply described as a record of digital events in the supply chain [Reyes et al., 2020]. However, in global supply chains, blockchain is much more than keeping digital records. The objective of blockchain in the supply chain is to improve trust in the system through visibility and transparency [van Hoek, 2019a], address the challenges of tamper-proof transaction records, facilitate

information sharing and synchronization, and enable smart contract execution [Helo & Hao, 2019]. Although blockchain adoption is seen as a disruptive technology, smart contracts would contribute to several supply chain management improvements, such as improved responsiveness, lead-time reduction, transaction-cost reduction, increased visibility, and more trust, security, and transparency in the global supply chain network [Queiroz *et al.*, 2019]. Its implementation can also provide accountability in interorganizational business collaboration [Kumar *et al.*, 2020], and it can play an important role in monitoring product quality and responding to product safety problems [Fan *et al.*, 2022].

However, these blockchain benefits are not achieved immediately, and several implementation barriers could impede blockchain adoption. One main issue with the blockchain is its transaction speed. At present, it can only process 4.6 transactions per second as compared to VISA that can process around 1700 transactions per second [Kenny, 2019]. Other barriers include user perception of risk and challenges of using blockchain, lack of understanding of the technology, the perceived value of the technology, trust, and security, implementation issues dealing with integration and compatibility, and system quality [van Hoek, 2019b].

There is confusion about the efficiency and effectiveness of blockchain performance. Cocco *et al.* [2017] defined three metrics, i.e., “economic efficiency,” “operational efficiency,” and “efficient service” concerning the banking industry. They concluded that blockchain could handle financial processes in a more efficient way than current systems. In a study of blockchain e-procurement systems, Thio-ac *et al.* [2019] analyzed the capability of the blockchain system’s “effectiveness” to hold transactional data with

complete decentralization and eliminate the growing number of fraud cases in companies and organizations.

Compounding the confusion concerning blockchain is the understanding of standards and the lack of information on blockchain implementation, which is referred to as the “grey space.” In a recent interview with William John Greaves (February 5, 2021), radio-frequency identification (RFID), Internet of Things (IoT), and Blockchain solutions architect at Lowry Solutions and Board of Directors of automatic identification and mobility (AIM), he defined the confusion concerning blockchain and the “grey space” as

While bitcoins was the precursor, blockchain in the supply chain is more than bitcoins. And blockchain today is where RFID was in 2002 and the bar code was in 1980. The problem we are seeing today with blockchain is first to develop models with utility and visibility to an ROI. Second is an understanding of the difference between the physical and the metaphysical in relation to the blockchain itself. Followed by understanding two specific standards: the platform is what blockchain is running on, and the process is what blockchain is running for. But the issue has to do with the grey space of blockchain; which is when companies are implementing blockchain but are not willing to communicate any information on their blockchain implementation.

Visich *et al.* [2009] observed similar issues around RFID. They identified a lack of information on successful implementations of RFID, which was leading to a perception that RFID was a failing technology. They attributed this to companies not wanting to share information about successful RFID deployments, thereby losing

competitive advantage. Moreover, unlike blockchain, the Auto-ID Labs and early mover industry partners played a major role in shaping RFID standards.

Insights from early blockchain adopters in the industry can be found in Verhoeven *et al.* [2018] and van Hoek [2019a, 2019c]. They offer guidance for the adoption of blockchain in the supply chain beyond the development of use cases that lead to the design of actual pilot studies. The frameworks consider questions for blockchain adoption (“what to adopt”) and identifying the right business opportunity (“where to start” and “what process or application are we doing”) for contributing to higher levels of global supply chain objectives (“what are the ROI expectations”).

Although these frameworks are useful for the pilot study and adoption decision, the reporting of actual implementation results is equally important to supply chain professionals and scholars. To reduce the grey space, there is a need to empirically study and report on actual implementations. Hence, we study the global supply chain performance based on efficiency and effectiveness resulting from blockchain technology (BCT).

This research study seeks to further empirical research by providing insights for scholars and practitioners on the evidence of blockchain implementation’s impacts on both supply chain efficiency and effectiveness performance. In this study, we review the recent relevant empirical literature on blockchains’ impact on supply chain efficiency and effectiveness performance. Next, we present the business process framework we use to organize the empirical evidence of supply chain performance improvements due to blockchain, and we classify the evidence according to the framework.

We then recommend nine propositions for future research. Finally, we conclude with a discussion of the research’s managerial and academic implications and our final thoughts.

2. LITERATURE REVIEW

The recent and explosive academic interest in blockchain has generated a rapidly growing body of blockchain and related literature. We limit our review to the literature on blockchain applications in supply chain management within the operations and supply chain management (OSCM) literature. We begin by reviewing the overview of blockchain, followed by a review of the blockchain literature review papers. From the literature, we contribute to this stream by utilizing the Mooney *et al.* [1996] framework to organize and demonstrate the empirical evidence of blockchain impacts on supply chain efficiency and effectiveness performance as described in Section 3.

2.1. Blockchain Overview Literature in Supply Chain Management Within the OSCM literature, there are a growing number of papers that provide an overview of BCT and its potential application benefits and barriers to implementation. We summarize a

collection of recent papers as a relevant representation of general overview studies concerning blockchain in supply chain management in Table 1.

Treiblmaier [2018] is among the first group of early blockchain researchers to provide an overview of blockchain and the potential implications for supply chain management. This article lays the foundation for further systematic and theory-based research to help academic researchers and industrial practitioners better understand blockchain and its potential implications for supply chain management (SCM).

The majority of the blockchain literature details the potential application benefits based on smart contracts for improving transparency and traceability. Blockchain technology is used to illustrate its potential application in business process management in Viriyasitavat *et al.* [2020]. They developed a blockchain approach to facilitate trust among businesses through smart contracts. Through their analysis, they concluded that blockchain was an ideal solution to replace experts and third-party authorities. By comparison, Barenji [2022] also considered the trust construct and developed a distributed-blockchain-based trust

system called “blocktrust” to address the trust issues in cloud manufacturing. This framework was developed using blocktrust and tested under different scenarios, resulting in a robust and credible system.

Blockchain can improve product safety and security, facilitate quality management, promote a reduction in illegal counterfeiting, contribute to sustainable supply chain management, and improve inventory control and replenishment [Cole *et al.*, 2019]. Blockchain can provide valid and tamper-proof manufacturing data, allowing for seamless traceability of that data to partners in the production process [Hasan & Starly, 2020]. Zhang *et al.* [2020] made recommendations for the application of blockchain in finance and economic service quality and considered BCT for solving two major problems of the digital economy: the flow of all digitalized assets and no-cost trust built anonymously. Sunny *et al.* [2020] used a proof of concept for a cold chain scenario using Microsoft Azure Blockchain Workbench, providing insights into the possibilities of using blockchain for traceability and supply chain transparency with technologies such as the IoT and smart contracts.

Although there is a general understanding that blockchain is a disruptive technology, much like past

Table 1. Summary of Blockchain Overview Literature in Supply Chain Management.	
Author(s)	Overview study emphasis
Treiblmaier [2018]	Potential implications of blockchain for SCM
Cole <i>et al.</i> [2019]	Identifies potential areas of application and provides an agenda for future research
Saberi <i>et al.</i> [2019b]	Application to sustainable supply chain management
Schmidt and Wagner [2019]	A transaction cost theory perspective
Fachrunnisa and Hussain [2020]	Blockchain-based human resource framework
Hasan and Starly [2020]	Decentralized cloud manufacturing as a service platform application
Sunny <i>et al.</i> [2020]	An overview of blockchain transparency and traceability solutions
Wamba and Queiroz [2022]	Proposes a multistage model of adoption, drawing on the diffusion of innovation theory, the resource-based view, dynamic capability, the technology adoption model, and the institutional theory
Zhang <i>et al.</i> [2020]	Challenges and countermeasures in finance and economics
Viriyasitavat <i>et al.</i> [2020]	An automated business process management solution is investigated to select and compose services in the open business environment
Barenji [2022]	BCT-based trust system called “Blocktrust” is proposed to address the trust problem of the cloud manufacturing
Kouhizadeh <i>et al.</i> [2021]	Theoretical exploration of adoption barriers
Mattila <i>et al.</i> [2021]	Blockchain-based approach for the deployment of product-centric information systems

disruptive technologies, there is a gap in the barriers to implementation research—but we can posit the similarities to past technology implementation. Wamba and Queiroz [2022] drew on the diffusion of innovations theory along with the technology adoption model to propose a multistage model adoption for blockchain. The authors proposed this model for a better understanding of blockchain diffusion across supply chains. Additional blockchain adoption barriers are explored in Kouhizadeh *et al.* [2021] using the technology–organization–environment framework and force field theories. The authors used literature streams on technology, organizational practices, and a comprehensive overview of barriers to examine sustainability in supply chains.

Other potential applications are being explored within the supply chain. Sustainability in the supply chain is a major factor of concern, as illustrated in Saberi *et al.* [2019b]. Blockchain's P2P verification helps to reduce the role of intermediaries in the supply chain network, and Schmidt and Wagner [2019] offered a transaction cost perspective on blockchain's application to the supply chain through transparent and valid transactions, potentially eliminating the need for intermediaries (third-party validation). Fachrunnisa and Hussain [2020] developed a human resource framework using blockchain

to help companies match their needs to the competencies of the available workforce by mapping corporate training centers to standardize the competencies presented in the training material. Mattila *et al.* [2021] used explorative design science and systematic combining to deploy a blockchain system for collecting product life-cycle data.

2.2. Blockchain Literature Review Papers

In our literature search, we identified recent OSCM literature review papers that present applications, benefits, and challenges, as well as future research directions. These are summarized in Table 2. The OSCM review papers focus on BCT potential with smart contracts and research the potential in supply chain and logistics management for increased supply chain visibility and traceability.

Finance, banking, and security sectors/industries where BCT has been used or where it has potential with smart contracts were identified by Gurtu and Johny [2019]. Queiroz *et al.* [2019] identified transportation, traditional supply chain management (distribution), and pharmaceutical supply chains among various significant segments for blockchain applications for increased supply chain visibility and traceability. The benefits of applying blockchain in supply chain management, including overall quality, reduction in costs,

shorter delivery times, reduction in risk, and an increase in trust, are summarized by Helo and Hao [2019].

From a managerial perspective, we are victims or beneficiaries of a multitude of converging technologies that are changing the way business is conducted in the global supply chain [Reyes *et al.*, 2020]. The potential applications and benefits of blockchain are shared in the OSCM literature review papers, but very few have focused on the challenges and risks. Hald and Kinra [2019] explored the potential enabling and constraining managerial implications of BCT for the supply chain, specifically indicating how blockchain could/would act and reach its projected effect, such as “blockchain facilitates,” “the blockchain revolution,” “transformative potential,” and “disruptive potential.” The challenges influencing blockchain adoption entail interorganizational trust, governance, transparency, immutability, exchange of information, selection of products, societal change, and behavior of the participants [Ghode *et al.*, 2020]. The potential operational risks of blockchain-integrated container shipping systems as a barrier to technology adoption specific to the container freight industry were investigated by Nguyen *et al.* [2021].

Recent systematic reviews offer thematic groupings for studying

Table 2. Summary of Recent Blockchain Literature Review Papers.

Author(s)	Literature review emphasis
Gurtu and Johny [2019]	Potential of BCT in SCM
Hald and Kinra [2019]	Identifies four enabling and three constraining blockchain identities to explain how the technology either “facilitates” or “impedes” SCM and performance
Helo and Hao [2019]	A classification of the applications of BCT in the scope of OSCM
Queiroz <i>et al.</i> [2019]	Blockchain and SCM integration
Wang <i>et al.</i> [2019a]	Focuses on the diffusion of BCT
Zhao <i>et al.</i> [2019]	Systematic literature network analysis
Ghode <i>et al.</i> [2020]	Blockchain adoption challenges
Hastig and Sodhi [2020]	Thematic analysis, critical success factors—cobalt mining and pharmaceuticals
Kasten [2020]	Systematic literature review of engineering and manufacturing applications
Lim <i>et al.</i> [2021]	A comprehensive analysis of themes, methodologies, and industries
Nguyen <i>et al.</i> [2021]	Investigates the potential operational risks of blockchain-integrated container shipping systems
Zutshi <i>et al.</i> [2021]	Presents a set of key value propositions that can complement the evolution of digital platforms and the collaborative economy

blockchain. Wang *et al.* [2019a] focused on the diffusion of BCT within supply chains and investigated how BCT is likely to influence future supply chain practices and policies. Although trust is the predominant factor driving the adoption, the value of blockchain lies in the four main areas of extended visibility and traceability, supply chain digitalization and disintermediation, improved data security, and smart contracts. Coupled with advanced information and communication technology and the IoT, blockchain can improve traceability, information security, manufacturing, and sustainable water management in the agrifood value chain [Zhao *et al.*, 2019]. They further identified the main challenges for adoption, including storage capacity and scalability, privacy leakage, high cost and regulation problems, throughput and latency issues, and lack of skills.

A thematic analysis by Hastig and Sodhi [2020] grouped the blockchain research into critical success factors identified for implementation as follows: companies' capabilities, collaboration, technology maturity, supply chain practices, leadership, and governance of the traceability efforts. Kasten [2020] categorized the use of blockchain in engineering and manufacturing activities into the

following three primary applications of blockchain: protect data validity; enhance inter- and intraorganizational communications; and increase the efficiency of the manufacturing process. Zutshi *et al.* [2021] identified key value propositions proposed by blockchain technologies applicable for integration with digital platforms and the collaborative economy by classifying the research into 12 themes, with 53% of the articles focused on cryptocurrencies, supply chain, the IoT, and finance.

Notably, the OSCM literature on blockchain-based supply chain research has attracted the attention of scholars from a global perspective. Lim *et al.* [2021] explored the value of blockchain in supply chains, the attraction of scholars to supply chain themes, the development of research methodologies and illustration types in adopting blockchain in supply chains, and the types of industries involved in blockchain-based supply chains.

3. EMPIRICAL EVIDENCE OF BLOCKCHAIN

Several frameworks have been suggested to conceptualize the business value of blockchain in the

supply chain. For example, van Hoek [2019b] reported that the interest in blockchain holds similarities to the interest surrounding RFID in 2004 and that there may be opportunities to leverage insights from the early RFID research to guide blockchain adoption. He explored how the Reyes *et al.* [2016] framework for RFID implementation may inform the consideration of blockchain in the supply chain. A two-stage approach is used to explore RFID implementation considerations from the Reyes *et al.* [2016] RFID implementation framework to blockchain by employing an initial exploration of managers interested in the blockchain using a focus group, a survey and in-depth exploration of three case companies pioneering blockchain.

Kim [2019] designed the *Blockchain Governance Game*, a network framework to prevent damage from an attacker. The Governance Game provides a new stochastic game framework to find the best strategies for preventing a network malfunction by attackers with the explicit equations for developing blockchain network security to avoid attacks on decentralized networks.

In our research, we use the business-process-oriented framework (see

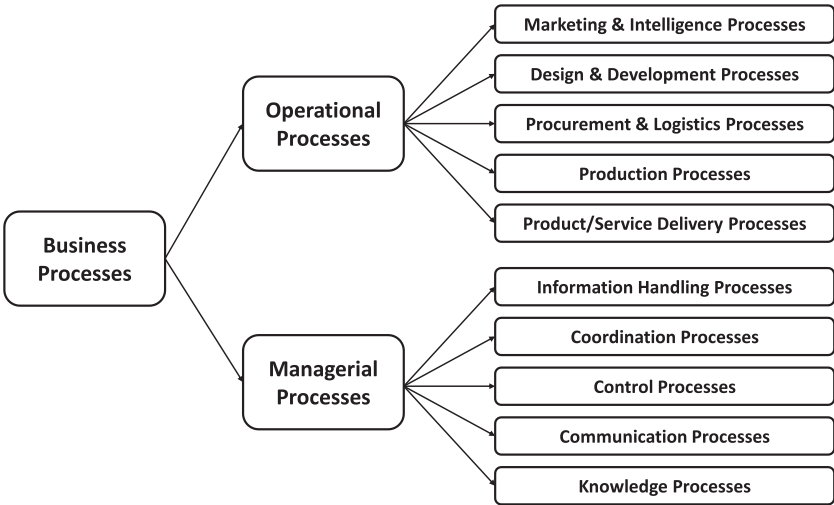


Figure 1. Typology of business processes (adopted from Mooney *et al.*, 1996).

Figure 1) proposed by Mooney *et al.* [1996] to organize our data. This framework has been used to study enterprise resource planning implementations ([Cotteleer and Bendoly, 2006]; [Karimi *et al.*, 2007]) and in-house versus outsourced information technology [Wang *et al.*, 2008]. Visich *et al.* [2009] applied the framework to investigate the actual empirical evidence of RFID impacts on supply chain performance. In the following sections, we present the framework, define the processes, and examine the three effects (automational, informational, and transformational). We then define the business value metrics and categorize the empirical evidence of blockchain impacts on supply chain efficiency and effectiveness performance in the framework.

3.1. Business-Process-Oriented Framework The business-process-oriented framework was first presented by Mooney *et al.* [1996] to facilitate the assessment of the business value of information technology. Figure 1 shows the typology of processes. Operational processes focus on efficiency and apply to those supply chain activities that are required to fulfill the actual physical work that makes up the value chain. Management processes are those activities that focus on effective decision-making associated with administration, allocation, and monitoring and control. These apply to those supply chain activities that are required to accomplish the actual

physical work that makes up the value chain.

Operational processes involve the execution of tasks, and management processes are facilitated through improved availability and communication of information. Since BCT is an information and communication technology that is being used in a variety of supply chain processes (and applications), we adapt this framework to explore the empirical evidence of blockchain impacts on supply chain efficiency and effectiveness performance.

For each business process, there are three linked effects on operational and management processes for value creation through BCT: automational, informational, and transformational. Automational effects correlate to the value derived from making the process more efficient. Informational effects refer to the ability of technology to gather, store, process, and distribute information. Transformational effects refer to the ability of technology to create process innovation and transformation. Table 3 summarizes the business value metrics for the processes and effects.

We use the following definitions and metrics for each effect:

Automational

- a) Operational processes seek to reduce labor costs, improve reliability and efficiency, and

reduce throughput and inventory costs.

- b) Management processes seek to reduce administrative expenses, better control of processes, easier reporting, and routinization.

Informational

- a) Operational processes seek to improve utilization, quality, and responsiveness, reduce waste, and improve operational flexibility.
- b) Management process metrics include improved effectiveness, decision quality, resource usage and empowerment, and higher levels of creativity.

Transformational

- a) Operational process metrics include product and service innovation and enhancement, reduction in cycle times, and a higher level of customer relationships.
- b) Management processes seek to improve competitive flexibility and capability and redesign organizational form.

We next report on our classification of the empirical evidence using the framework discussed. In our review of the empirical evidence of blockchain in the supply chain, we follow the work of Visich *et al.* [2009] and report on those papers that are based on actual results reported from a pilot study or an implementation. We purposely do not include hypothetical

Table 3. Business Value Matrix (Adopted from Mooney *et al.*, 1996).

	Automational	Informational	Transformational
Operational (focus on efficiency)	Labor cost Reliability Efficiency Inventory costs Throughput	Utilization Quality Responsiveness Waste Operational flexibility	Product and service innovation Cycle times Customer relationships
Management (focus on effective decision-making)	Administrative expense Control Reporting Routinization	Effectiveness Decision quality Resource usage Empowerment Creativity	Competitive flexibility Competitive capability Organizational form

deployments or estimated benefits. In our classification, we focus on efficiency to identify operational processes and on effective decision-making for managerial processes. We use the business value metrics for each process effect as a guideline to organize the data.

3.2. Operational Processes

Empirical Evidence A total of 33 operational business values for blockchain were identified, as shown in Table 4. The largest number (12 examples) were categorized as automational, 11 were identified as informational, and 10 as

transformational. Table 4 shows that automational effects of blockchain on operational processes include efficiencies resulting from Smart Contracts, more efficient and reliable traceability and transparency, and increased throughput as the result of automated processes and data dissemination.

One of the major automational effects is intelligent contract technology, sometimes called Smart Contracts. Smart Contracts hold the promise of overcoming the trust issue in supply chains by directly connecting buyers and suppliers using algorithms that

automatically identify the most efficient or effective solutions ([Zhang *et al.*, 2019]; [Hasan and Starly, 2020]), all while ameliorating the trust issue, which is the greatest obstacle to distributed supply chains, such as cloud manufacturing ([Zhu *et al.*, 2020]; [Barenji, 2022]). The shift to “cloud manufacturing” or “Cloud Manufacturing-as-a-Service” provides strong cases for blockchain implementation since they depend heavily on distributed system infrastructure while also requiring collaboration, the integrity of data, and transparency [Hasan & Starly, 2020]. Other supply chain services,

Table 4. Dimensions of Blockchain Operational Business Value.

Automational Efficiency	Secure data dissemination, personal health information in one view [Rahmadika & Rhee, 2018] Smart Contract with Industrial IoT sensors on shipping containers [Hasan <i>et al.</i> , 2019] Utilization metrics are developed for the Industrial IoT and blockchain utilization with RFID and the combined effect on supply chain transparency [Zelbst <i>et al.</i> , 2020] Smart Contract blockchain network that directly connects firms on a Cloud Manufacturing-as-a-Service platform [Hasan & Starly, 2020] Game analysis between supply chain factoring and blockchain technology for automated intelligent contract technology [Zheng <i>et al.</i> , 2022] Assesses pricing policy in the context of cloud manufacturing [Zhu <i>et al.</i> , 2020]
Reliability	Proposes an application design for blockchain-enabled aircraft leasing asset management that fulfills both regulatory and business concerns [Kuhle <i>et al.</i> , 2021] Application of new “Blocktrust” technology to cloud manufacturing [Barenji, 2022] Assesses how BCT addresses both confidentiality and transparency [Xu <i>et al.</i> , 2021]
Throughput	Develops and analyzes a blockchain-algorithm-guiding manufacturing service collaboration to choose optimal task providers [Zhang <i>et al.</i> , 2019] Analyzes the effect of P2P-data-sharing network on supply chain transparency by creating a digital twin of a product [Huang <i>et al.</i> , 2020] Proposes a business process management automated solution [Viriyasitavat <i>et al.</i> , 2020]
Informational Utilization	Real data from a case study of blockchain efficiencies for a customer order process; it reduces the number of operations, reduces the average time of orders in the system, reduces workload, shows traceability of orders, and improves visibility to various supply chain participants [Martinez <i>et al.</i> , 2019] Analytically demonstrates blockchain adjudicating a revenue-sharing contract in fashion [Choi, 2020] An event study to quantify the financial effects of the COVID-19 pandemic [Xiong <i>et al.</i> , 2021]
Quality	Proposes a traceability framework using BCT in textile and clothing [Agrawal <i>et al.</i> , 2021] Case study used for designing a blockchain system to prevent counterfeiting in the wine supply chain [Danese <i>et al.</i> , 2021] Develops model for centralized access to patient medical records across all of India [Dhagarra <i>et al.</i> , 2019] Propose a traceability model for olive oil [Guido <i>et al.</i> , 2020]
Responsiveness	Analysis of efficiencies and resilience in blockchain-enabled supply chains [Lohmer & Lasch, 2020] Design and implement a blockchain-enabled data-sharing marketplace [Wang <i>et al.</i> , 2021] Insights from a humanitarian supply blockchain pilot project [Baharmand <i>et al.</i> , 2021] Develops theory for requirements of an efficient blockchain-based supply chain [Azzi <i>et al.</i> , 2019]
Waste Transformational Product and service innovation	Present a framework for the theoretical impact of blockchain on supply chain management [Treiblmaier, 2018] Conceptual answer to end-to-end digital twin in the aircraft industry [Mandolla <i>et al.</i> , 2019] A platform for new research that identifies the transformational capabilities of the blockchain [Cole <i>et al.</i> , 2019] Model of BCT adoption that is compared in India versus the United States [Wamba & Queiroz, 2022] An expert interview study of potential and barriers to blockchain supply chain implementation [Lohmer & Lasch, 2020] Presents opportunities for the application of blockchain in the military and aerospace industries [Ahmad <i>et al.</i> , 2021]
Cycle times	A permissionless blockchain system for collecting product life-cycle data is conceptualized, demonstrated, and evaluated by experts [Mattila <i>et al.</i> , 2021]
Customer relationships	Develop a framework to build a robust, data-driven agrifood supply chain [Kamble <i>et al.</i> , 2019] A prototype system is developed that allows universal plug-and-play of heterogeneous supply chain resources [Li <i>et al.</i> , 2019] Use blockchain to develop energy communities using Smart Contracts [Petri <i>et al.</i> , 2020]

such as factoring [Zheng *et al.*, 2022] and business process management [Viriyasitavat *et al.*, 2020], also benefit from blockchain's unique ability to provide both transparency and confidentiality [Xu *et al.*, 2021].

In addition to reliably connecting distributed manufacturing and service networks, blockchain implementations of Smart Contracts can automate much of the tedious work of automatically tracking movement and releasing funds and information, greatly improving overall supply chain efficiency and reliability [Zheng *et al.*, 2022]. Smart Contracts, in combination with the Industrial IoT and sensors, such as RFID, promise to automate the connection between logistics and contracts. Interestingly, one pre-COVID paper identified the business case of a vaccine supply chain since they require constant monitoring of the shipping environment, such as temperature and humidity in combination with other shipment tracking information [Hasan *et al.*, 2019]. The combination of sensors, blockchain, and Smart Contracts has been demonstrated to facilitate the regulatory and compliance-intensive aircraft leasing management process [Kuhle *et al.*, 2021]. Healthcare is another industry characterized by high levels of regulations and compliance, and blockchain exhibits traits of secure data dissemination while providing a common set of patient records across all providers that would greatly improve the efficiency and reliability of patient data access [Rahmadika & Rhee, 2018].

The automational effects of blockchain enable supply chains to facilitate traceability and life-cycle analysis. Secure and efficient traceability is key to supply chain traceability in particular, and utilization metrics have been developed for the Industrial IoT and blockchain utilization with RFID to provide traceability [Zelbst *et al.*,

2020]. Analysis indicates that blockchain's data integrity capabilities to make digital twins enabled by blockchain-powered Smart Contracts are useful for life-cycle analysis via a P2P network that enhances data sharing efficiency among supply chain participants [Huang *et al.*, 2020]. Blockchain's ability to assure data integrity also makes it useful in the context of assessing pricing policy for distributed environments such as cloud manufacturing ([Zhu *et al.*, 2020]; [Barenji, 2022]).

The evidence of informational effects from blockchain overlaps to some degree with the automational effects yet is distinguished by the limited use of intelligent technology features. The most common uses of information effects were for quality and responsiveness. Quality applications of blockchain have been analyzed based on blockchain's traceability capacity. Traceability results from the inherent informational efficiencies of a blockchain-based supply chain [Azzi *et al.*, 2019]. Traceability can be used to monitor quality, such as for the olive oil industry [Guido *et al.*, 2020], and can enhance the responsiveness of the textile and clothing industry seeking sustainability [Agrawal *et al.*, 2021]. Regarding quality, one large-scale potential implementation is to create centralized access to patients' medical records across all of India, with the potential to improve accessibility and equity of healthcare across the country [Dhagarra *et al.*, 2019].

Informational effects based on utilization include a presentation of real data from a case study of the blockchain efficiencies for a customer order process [Martinez *et al.*, 2019]. Results indicated that blockchain reduced the number of operations, the average time of orders in the system, and workload; showed traceability of orders; and improved visibility to various supply chain participants. The informational effects

of blockchain were demonstrated to empower the adjudication of a revenue-sharing contract in the fashion industry [Choi, 2020]. Case studies have been used to explore the informational effects of improving visibility and transparency. Baharmand *et al.* [2021] provided main drivers and barriers insights from a humanitarian supply blockchain pilot project study. Danese *et al.* [2021] contributed with an investigation of how blockchain systems can prevent counterfeiting in the wine supply chains, and, in a similar approach, Xiong *et al.* [2021] conducted an event study on the role of blockchain during the COVID-19 pandemic.

The transformational effects of blockchain mainly resulted primarily from product and service innovation, especially the reenvisioning of supply chain management. Specific industry applications were analyzed for the military and aerospace industries [Ahmad *et al.*, 2021], with one study implementing transformation in the aircraft industry via an end-to-end digital twin. Other studies consulted with experts to identify transformational potential in blockchain-enabled supply chains [Cole *et al.*, 2019], sometimes with a specific eye toward both potential and barriers [Lohmer & Lasch, 2020]. One study focused on developing and comparing a model of supply chain BCT adoption between India and the United States [Wamba *et al.*, 2020], whereas another developed a framework for the transformational impacts of blockchain on the supply chain [Treiblmaier, 2018].

Several studies focused on the transformational possibilities of blockchain in supply chain relationships. These studies generally focused on blockchain's potential to leverage the combinatorial possibilities inherent in the extended supply chain. One study developed a prototype system that

allows university plug-and-play of heterogeneous supply chain resources [Li *et al.*, 2019]. Another study implemented blockchain-powered Smart Contracts to develop energy communities to foster innovative communities of sustainable energy users [Petri *et al.*, 2020]. The agrifood industry also received attention in the form of a framework to build a robust, data-driven supply chain [Kamble *et al.*, 2019].

One detailed study enlisted experts who conceptualized, demonstrated, and evaluated a permissionless blockchain system for collecting product life-cycle data [Mattila *et al.*, 2021]. In all the transformational studies, blockchain-enabled supply chains were catalysts to reimagine the efficiencies and possibilities in

modern supply chains, especially about innovation resulting from cocreating innovations with customers.

In sum, although many research studies have focused on prototyping and assessing expert opinion, blockchain demonstrates the ability to revolutionize supply chain operations in a way that will redefine business value. The technology exists, yet examples of operational deployments are still in the nascent period.

3.3. Managerial Processes Empirical Evidence

Table 5 lists 21 examples of empirical evidence of blockchain managerial business value distributed as six automational, six informational, and nine transformational effects. To date, research work into managerial

processes has focused somewhat more on transformation effects, especially with regard to looking simultaneously at blockchain's benefits in conjunction with its diffusion or stage of adoption.

A theme among automational effects is an exploration of barriers to adoption and environmental conditions favorable to blockchain implementation, albeit in different contexts. The most strategic treatment applied game theory to identify the conditions in which blockchain is worth or not worth implementing while also analyzing which applications would be more operationally convenient and economically appealing [de Giovanni, 2020]. Another study introduced four BCT adoption barriers in the context of sustainable supply chains [Saberi

Table 5. Dimensions of Blockchain Managerial Business Value.

<i>Automational</i>	
Administrative expense	Introduce four BCT adoption barriers, with a focus on sustainable supply chains [Saberi <i>et al.</i> , 2019b] A game-theoretic model identifies conditions in which blockchain is not worth implementing and applications that are more operationally convenient and economically appealing [de Giovanni, 2020]
Control	Blockchain influence on supply chain relationships, especially regarding transaction costs and governance decisions [Schmidt & Wagner, 2019] Experiment testing new blockchain data structure finds times to construct secure data at centralized versus decentralized supply chain nodes [Suhail <i>et al.</i> , 2020]
Reporting	Digital asset valuation and risk assessment in e-commerce logistics [Harish <i>et al.</i> , 2021]
Routinization	Systematic analysis of BCT and its application in finance [Zhang <i>et al.</i> , 2020]
<i>Informational</i>	
Effectiveness	Presents the blockchain governance game to find the best strategies for preventing a network dysfunction [Kim, 2019] Describes the effectiveness of blockchain at enhancing trust and visibility in food supply chains utilizing reliable data and fully visible exchanges [Rogerson & Parry, 2020] A sample of 143 Malaysian Halal food and beverage manufacturers explains 73% of the variance in intention to participate in a model of blockchain adoption [Hew <i>et al.</i> , 2020]
Decision quality	Models effectiveness of blockchain for combatting deceptive and nondeceptive counterfeiting [Pun <i>et al.</i> , 2021] Analyzes activity- and artifact-centric modelling paradigms to aid analysts in producing better process models for communication purposes [Milani <i>et al.</i> , 2021]
Empowerment	Develops a blockchain-based human resources framework to match company needs to workforce competencies [Fachrunnisa & Hussain, 2020]
<i>Transformational</i>	
Competitive flexibility	Compares and contrasts academic and practitioner perspectives on benefits and barriers to adopting blockchain using the Decision-Making Trial and Evaluation Laboratory tool [Kouhizadeh <i>et al.</i> , 2021]
Competitive capability	Review of existing use cases for opportunities for sustainability from blockchain use in the energy industry, including in developing countries [Bürer <i>et al.</i> , 2019] Manufacturing service collaboration based on the Industrial Internet platform [Zhang <i>et al.</i> , 2019]
Organizational form	Presents a four-layered framework of blockchain with a physical internet grounded in an industry case that guides research and practitioner adoption [Meyer <i>et al.</i> , 2019] In-depth interviews with experts regarding how blockchain will transform supply chains [Wang <i>et al.</i> , 2019b] Develops and validates a blockchain-enabled circular supply chain for fast fashion [Wang <i>et al.</i> , 2020] Grounded theory across multiple cases examines the practices, gaps, potential tensions, and critical reflections inherent in blockchain in the circular economy [Kouhizadeh <i>et al.</i> , 2021] A survey of 173 respondents benchmarks current practice with blockchain in the supply chain, describing motivators and barriers [Saberi <i>et al.</i> , 2019a] A survey in India and the United States validates a multistage model of blockchain technology adoption that explains all stages of blockchain diffusion [Wamba & Quieroz, 2022]

et al., 2019b]. These studies focused on blockchain's ability to reduce administrative expenses.

One study focused on the control in the context of governance decisions while also considering transaction costs [Schmidt & Wagner, 2019]. A transaction cost not often considered is time to encrypt data, which is a concern for blockchain applied to the context of real-time supply chain management; the study analyzed the time to construct secure data at centralized versus decentralized supply chain nodes, finding the delays nonprejudicial to managing supply chains [Suhail *et al.*, 2020]. The last couple of studies focused on reporting and routinization of assets in the supply chain, with one valuing digital assets and risk assessment in e-commerce logistics [Harish *et al.*, 2021], whereas Zhang *et al.* [2020] presented a systematic analysis of BCT and its application in finance.

The biggest group of empirical work on informational effects focused on blockchain's ability to enhance supply chain effectiveness. Two studies elaborated on blockchain's ability to improve managerial processes for food quality, one focusing on halal food and beverages in Malaysia [Hew *et al.*, 2020] and the other more focused on baby food, with findings generalizable to other contexts [Rogerson & Parry, 2020]. In addition to describing the enhanced effectiveness for managerial business value, both studies presented models to inform blockchain adoption decisions. One promising informational effect study looked at improving decision quality in the context of counterfeiting, assessing blockchain's efficacy for combatting both deceptive and nondeceptive counterfeiting [Pun *et al.*, 2021]. Another study informed the blockchain decision regarding which modeling paradigm better facilitates process models for communication purposes [Milani *et al.*, 2021]. The

last informational effect study focused on empowerment via a human resources framework that matches company needs to workforce competencies [Fachrunnisa & Hussain, 2020].

Transformational effects of blockchain manifested as competitive flexibility especially with regard to technology, organizational practices, and sustainability, with both academics and practitioners agreeing that supply chain and technological barriers are most critical to blockchain implementation [Kouhizadeh *et al.*, 2021]. Transformational effects appeared in the form of the competitive capability to achieve sustainability in B rer *et al.*'s [2019] evaluation of the benefits and barriers to blockchain implementation in the energy industry, including in developing countries, whereas Zhang *et al.* [2019] assessed manufacturing service collaboration using an industrial internet platform.

Transformational effects on organizational form appeared in the work of Meyer *et al.* [2019], who presented a four-layered framework of blockchain with physical internet grounded in an industry case that guides research and practitioner adoption. Wang *et al.* [2019b] conducted in-depth interviews with experts regarding how blockchain will transform supply chains. Wang *et al.* [2020] developed and validated a blockchain-enabled circular supply chain for fast fashion. Research work into the circular economy also appears in a grounded theory assessment of multiple cases, revealing prevalent practices, gaps, potential tensions, and critical reflections [Kouhizadeh *et al.*, 2021]. A complementary study of motivators and barriers to blockchain implementation in supply chains was accomplished with a survey of 173 respondents benchmarking current practice [Saber *et al.*, 2019a]. Useful to companies contemplating their

state of blockchain adoption is a study by Wamba and Quieroz [2022] that surveyed supply chain managers in India and the United States to validate a multistage model of BCT adoption that provides an explanation at all stages of blockchain diffusion.

In total, the managerial business value potential offered by blockchain implementation in supply chains is both broad and deep. Implementations have a focus on sustainable supply chains that revolutionize trust in supply chain relationships with data and technology. The research work is just starting to consider factors related to human talent and managerial processes, with a special focus on understanding the unique value proposition of logistics and supply-chain-related assets in the forthcoming digital era.

3.4. Discussion of Findings and Propositions Our analysis of the operational and managerial business value from blockchain implementation indicates no dominant effect across the automational, informational, and transformational categories. Thirty-three papers were identified as operational, whereas 21 were managerial. These results differ from the work of Visich *et al.* [2009], who found that for RFID, the major effects were automational for operational business value and informational for managerial business value, with a large majority of the evidence indicating operational business value.

Mooney *et al.* [1996] proposed a three-stage model for the impact of information technology on business value. In this model, first-stage effects for operational business value are automational and for managerial business value, they are informational. The second-stage effect for operational is informational, whereas for managerial, they are automational. Third-stage effects are

transformational for both operational and managerial.

Our results for blockchain do not support the Mooney *et al.* [1996] framework since the number of papers in each business value classification is fairly close, with a range of four for operational and three for managerial. For operational, the largest number of papers was found for automational (12), followed by transformational (10), and then informational (8). For managerial, there were nine transformational papers, whereas automational and informational each had six papers. However, some subcategories had a larger number of papers than other subcategories. Operational–automational had six papers on efficiency, operational–transformational had six papers for product and service innovation, and managerial–transformational had six papers for organizational form.

Unlike RFID, which began commercial use in the 1980s in the railroad and trucking industries [Landt, 2001] and then in manufacturing [Stall, 1993] before widespread supply chain use, blockchain is a relatively immature technology. In addition, although early RFID implementations benefited from numerous pilot tests [Visich *et al.*, 2009], we are not aware of any such large-scale pilot testing with blockchain. Based on our analysis of the empirical blockchain implementations and using the work of Visich *et al.* [2009] as a guide, we make the following propositions regarding BCT in the supply chain. These propositions can be tested or modified in future blockchain research.

The first-order effects of blockchain will be automational effects on operational process efficiency by reducing labor costs, automating inbound logistics associated with receiving inventory, and reducing

throughput time. Automational effects on the managerial processes are linked with effective decision-making on routine processes, such as inventory planning and control, reporting, and administrative control of accounts payable for inventory received. The automational effects of blockchain on operational and managerial processes include more reliable traceability and transparency, increased throughput, and overcoming the trust issue in global supply chains resulting from Smart Contracts. The primary use of Smart Contracts in the blockchain holds the promise of overcoming the trust issue that is common across global supply chains. Smart Contracts will provide more reliable traceability and transparency of the material flow in the global supply chain channels. The following propositions encapsulate these findings' automational effects.

- P1: Blockchain will enable automational effects for operational processes first.
- P2: The effects of blockchain on operational processes will enable higher levels of efficiency through automation.
- P3: Blockchain will facilitate operational and managerial processes when it is used to improve traceability in the supply chain to product input origins.

The second-order effects of blockchain will be informational effects on operational process efficiency with the availability of rich information across the blockchain network. The central focus of this stage will be a better understanding of the information when data can be collected and disseminated from automational effects. With the operational focus on efficiency, the informational effects blockchain should be able to improve are the utilization of resources, operational flexibility, responsiveness, and waste reduction based on the rich information availability. The informational effects of blockchain on

managerial processes include effective decision-making based on the information generated from the blockchain network. Effectiveness and decision quality in planning and control can be linked with forecasting, inventory replenishment, and fulfillment. Based on this evidence, the following propositions for the informational effects are offered.

- P4: For operational processes, blockchain will facilitate informational effects when data can be collected and disseminated from automational effects.
- P5: The effects of blockchain on operational processes will enable product and service innovation that is transformational.
- P6: For managerial processes, blockchain will facilitate informational effects when data can be collected and disseminated from automational effects.

The third-order effects of blockchain will be the transformational effects of blockchain on operational and managerial processes driven by process innovation, leading to global supply chain redesign to achieve an increase in competitive advantage. Managerially, the effectiveness of decision-making from accurate and real-time data should transform the global supply chain dynamics. As a result, organizations should then focus on dynamic scheduling across the blockchain network, thus providing a competitive advantage for scheduling flexible capabilities. This would then increase global supply chain efficiencies in lead time reduction and improved cycle times—leading to improved supplier and customer relationships in the blockchain network. We, therefore, make the following propositions for the transformational effects.

- P7: Blockchain will enable transformational effects for managerial processes first.

P8: The effects of blockchain on managerial processes will enable new organizational forms that are transformational.

P9: For operational processes, blockchain will facilitate transformational effects when the process is reengineered to significantly reduce production or supply chain cycle times.

4. RESEARCH IMPLICATIONS AND CONCLUSIONS

The empirical evidence presented in this research illustrates the impact of blockchain in the supply chain on operational business value and managerial business value, and it has attempted to develop a better understanding of the stages blockchain will go through as it matures. Based on the evidence we have identified, we do not believe blockchain will follow the stage model presented by Mooney *et al.* [1996] until the following conditions occur. First, industry-standard protocols that allow for a secure, open interface need to be developed so that blockchain is easily accessible to all verified partners in a supply chain. This will allow all members of a supply chain access to information that they can use to enhance competitive performance. Second, more pilot studies need to be conducted in a wide variety of processes to identify best implementation practices and how to overcome barriers. These pilot studies should be done for standard business processes as well as innovative business processes, and

the knowledge gained should be shared with a wider audience of practitioners and academics. Third, our research study did not find any large-scale, end-to-end applications of blockchain among multiple supply chain partners. Blockchain implementations need to take on a more strategic nature, where multiple partners and processes along the supply chain contribute to and share the data within blockchain. As more organizations implement blockchain, the operational and business value of blockchain should increase. Fourth, more research work needs to be conducted on the costs and return on investment of blockchain implementation and how those costs can be shared by supply chain partners. Although there is a higher value in shared information, not all partners will benefit equally from that information; hence, how much each entity pays is a critical issue.

Our research study has implications for academics and practitioners. Academics can use the data presented to develop and test more complex blockchain-enabled supply chain models. They can also test the propositions through survey research and or field research. Practitioners can benchmark successful implementations discussed in this research to identify opportunities where blockchain can be deployed within their company and, more importantly, across their supply chains. They can use this information to develop a business case for blockchain and better estimate the financial implications.

There are several limitations of our research. First, the empirical evidence we report used secondary sources of information, primarily peer-reviewed journal publications. None of these business value benefits of blockchain were collected directly by the authors. Our second limitation is a lack of consistency in the definitions of the subcategories. For example, quality was a subcategory within operational–informational and of the three papers identified, two were focused on traceability quality, whereas the other focused on patient record quality. The third limitation is the lack of quantifiable data in the research, which clouds the true impact of blockchain in the supply chain. The fourth and final limitation of this research is our focus on English-language articles. Due to the global interest in blockchain, we are likely to have missed relevant articles published in other languages.

To the best of our knowledge, this is the first research to collect and organize empirical evidence on blockchain implementation. We have utilized the Mooney *et al.* [1996] framework to organize this evidence into operational and managerial business value processes and the effects: automational, informational, and transformational. In addition, we have identified possible reasons why blockchain is not following the Mooney *et al.* [1996] stage model, and we have offered suggestions on how to facilitate blockchain deployment.

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